

Week 7 – Multi-Language Integration & Demonstration

Summary:

Week 7 focused on **integrating all backend features** and preparing demonstration scripts for showcasing multi-language compiler capabilities.

Key Activities:

- Combined **Python, C, C++, Java execution endpoints** into a single unified backend service.
- Created **sample programs** demonstrating input/output, computational tasks, and graphical outputs.
- Verified integration inside Docker container with persistent volume mapping.
- Tested overall workflow for multiple simultaneous users.

Technologies Used: Python, FastAPI, Docker, GCC, g++, JDK, Matplotlib

Outcome:

Backend is now a **fully integrated multi-language compiler system**, ready for demonstration and future frontend integration.

Flowchart:

